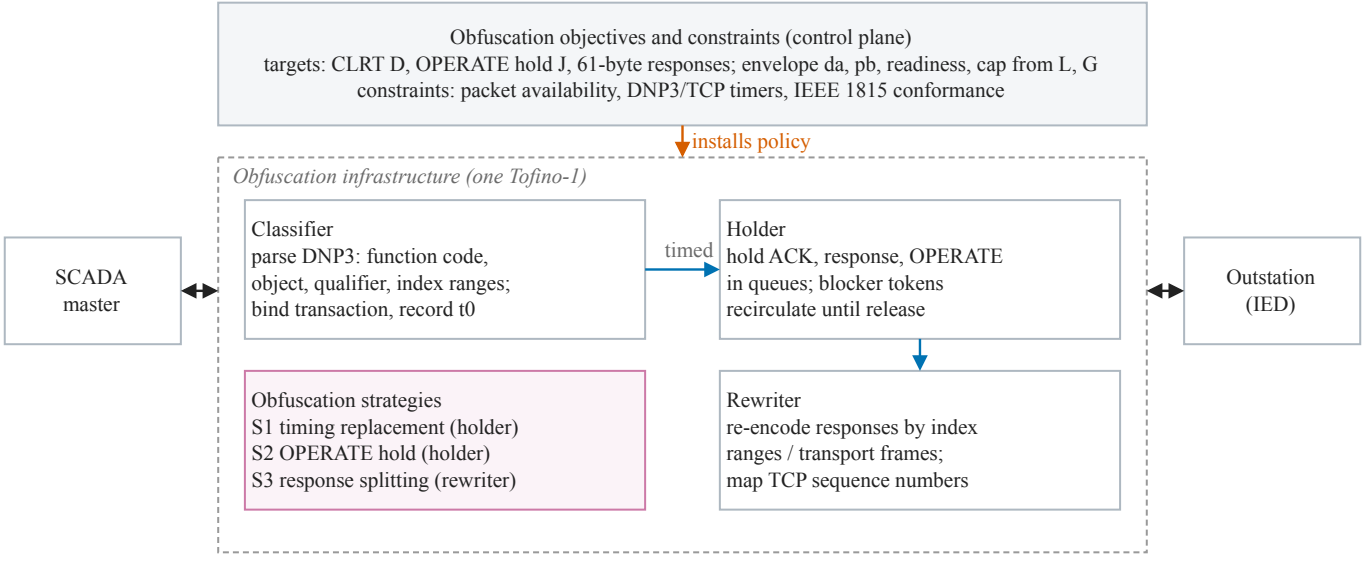
Obfuscation objectives and constraints (control plane)
targets: CLRT D, OPERATE hold J, 61-byte responses; envelope da, pb, readiness, cap from L, G
constraints: packet availability, DNP3/TCP timers, IEEE 1815 conformance
installs policy
Obfuscation infrastructure (one Tofino-1)
SCADA master
Classifier
parse DNP3: function code, object, qualifier, index ranges; bind transaction, record t0
timed
Holder
hold ACK, response, OPERATE in queues; blocker tokens
recirculate until release
Outstation (IED)
Obfuscation strategies
S1 timing replacement (holder)
S2 OPERATE hold (holder)
S3 response splitting (rewriter)
Rewriter
re-encode responses by index ranges / transport frames;
map TCP sequence numbers